

DSA STUDY BLUEPRINT SERIES

69. Trapping Rainwater Problem Optimal Solution

Calculating Trapped Rainwater Volume

Section 08 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Rainwater:** Water trapped at index i is $\min(\text{maxLeft}, \text{maxRight}) - \text{height}[i]$.
- Can be computed using prefix arrays, monotonic stacks, or two pointers.

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
int trap(vector& height) {  
    int l = 0, r = height.size() - 1, ans = 0;  
    int leftMax = 0, rightMax = 0;  
    while (l < r) {  
        if (height[l] < height[r]) {  
            if (height[l] >= leftMax) leftMax = height[l];  
            else ans += leftMax - height[l]; l++;  
        } else {  
            if (height[r] >= rightMax) rightMax = height[r];  
            else ans += rightMax - height[r]; r--;  
        }  
    }  
    return ans;  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Two Pointer Search	$O(N)$ time, $O(1)$ space

Key Practices & Gotchas

- **Important:** The two-pointer technique solves the problem in a single pass with $O(1)$ auxiliary space.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace monotonic stack modifications for next greater element on `[4, 5, 2]` :

Active Element	Stack State (Indices)	Pop comparison	Result output
4	[0] (value 4)	N/A	Waiting for next greater
5	[1] (value 5)	5 > 4 (pop index 0)	Next Greater of 4 is 5

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Monotonic properties:** Stacks process elements strictly increasing or decreasing, optimizing nested lookups.
- **Min Stack details:** Track current minimum bounds by storing elements as encoded difference values.
- **Related LeetCode Problems:** #155 Min Stack, #84 Largest Rectangle in Histogram, #42 Trapping Rain Water.